

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0749

Arcogen 138 kV
July 31, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0749. The Interconnection Customer proposes a 204.9 MW solar photovoltaic facility interconnecting at the Arcogen 138 kV substation in the Virginia area. The request was received on February 21, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 298 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$24,220,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Tasley 345/138kV TX: 153.9% loading (344 MVA) following the outage of the Southtown to Commonwealth Chesapeake 345.0 kV; DFAX = 66.3%.

The Vienna Operations 345 kV/1 kV Transformer is overloaded to 146.1% of its 552 MVA emergency rating for the loss of the Vienna Operations to Easton 2 345.0 kV. The project contributes 49.4% DFAX (101.3 MW) to this overload.

For the loss of Vienna Operations to Easton 2 Substation 345.0 kV Line, the Sunset to Saddlebrooke Ranch 138.0 kV loads to 143.9% of its 512 MVA rating. Project DFAX: 42.0%.

Chemolite to Randall County 138.0 kV L: 138.5% loading (230 MVA) following the outage of the Easton 2 to Vienna Operations 345.0 kV; DFAX = 15.3%.

The Newmark to Ess Lewes 138.0 kV L is overloaded to 130.2% of its 368 MVA emergency rating for the loss of the Vienna Operations Substation to Easton 2 345.0 kV Line. The project contributes 25.0% DFAX (51.3 MW) to this overload.

For the loss of Southtown to Vienna Operations 345.0 kV, the Galvanizing to Santeetlah Substation 138.0 kV line loads to 127.1% of its 520 MVA rating. Project DFAX: 22.0%.

The Santeetlah to Sunset 138.0 kV is overloaded to 126.6% of its 527 MVA emergency rating for the loss of the Southtown to Vienna Operations 345.0 kV. The project contributes 22.0% DFAX (45.1 MW) to this overload.

Cady to Easton 2 345.0 kV Line: 125.9% loading (1,185 MVA) following the outage of the Navy Mole to Holston Dam 138.0 kV; DFAX = 64.7%.

For the loss of Easton 2 to Vienna Operations 345.0 kV Line, the Ripple Substation 345 kV/138 kV XFMR loads to 125.4% of its 433 MVA rating. Project DFAX: 16.3%.

Sandburg to Jarratt 138.0 kV: 122.7% loading (542 MVA) following the outage of the Vienna Operations to Easton 2 345.0 kV; DFAX = 27.2%.

The Sherwood to Sandburg 138.0 kV is overloaded to 121.9% of its 554 MVA emergency rating for the loss of the Easton 2 to Vienna Operations 345.0 kV. The project contributes 27.2% DFAX (55.6 MW) to this overload.

Hitchland to Sherwood 138.0 kV Line: 121.5% loading (561 MVA) following the outage of the Vienna Operations to Easton 2 345.0 kV; DFAX = 27.2%.

The North Lake to Hitchland 138.0 kV is overloaded to 119.7% of its 591 MVA emergency rating for the loss of the Vienna Operations to Easton 2 345.0 kV. The project contributes 27.2% DFAX (55.6 MW) to this overload.

For the loss of Vienna Operations to Easton 2 345.0 kV, the Kiwi to North Lake 138.0 kV loads to 118.8% of its 528 MVA rating. Project DFAX: 24.7%.

The Sunset to St Bonifacius 138.0 kV is overloaded to 115.9% of its 1,064 MVA emergency rating for the loss of the Southtown to Vienna Operations 345.0 kV. The project contributes 54.5% DFAX (111.6 MW) to this overload.

W -Station to St Bonifacius 138.0 kV line: 115.3% loading (1,089 MVA) following the outage of the Southtown to Vienna Operations 345.0 kV L; DFAX = 54.5%.

For the loss of Southtown to Vienna Operations 345.0 kV, the Saddlebrooke Ranch to Hansford 138.0 kV loads to 113.8% of its 376 MVA rating. Project DFAX: 7.2%.

The Southtown to Quince Orchard 138.0 kV is overloaded to 113.3% of its 236 MVA emergency rating for the loss of the Commonwealth Chesapeake to Southtown 345.0 kV L. The project contributes 8.4% DFAX (17.2 MW) to this overload.

For the loss of Commonwealth Chesapeake to Southtown 345.0 kV line, the Prunedale to Dane 138.0 kV loads to 112.6% of its 226 MVA rating. Project DFAX: 6.2%.

Mesquite to Ripple 138.0 kV: 112.6% loading (824 MVA) following the outage of the Vienna Operations to Southtown 345.0 kV; DFAX = 45.5%.

The Mesquite to Arroy Tap 138.0 kV is overloaded to 111.5% of its 859 MVA emergency rating for the loss of the Vienna Operations to Southtown 345.0 kV. The project contributes 45.5% DFAX (93.2 MW) to this overload.

For the loss of Easton 2 to Cady 345.0 kV, the Bristol to Limon Substation 138.0 kV loads to 111.4% of its 258 MVA rating. Project DFAX: 11.5%.

Grasonville Substation to Wye Mills Substation 138.0 kV: 111.3% loading (203 MVA) following the outage of the Cady to Easton 2 345.0 kV; DFAX = 6.5%.

Ocean City Substation to Reidland 138.0 kV: 111.0% loading (878 MVA) following the outage of the Southtown to Vienna Operations 345.0 kV L; DFAX = 45.5%.

Arroyo Tap to Reidland 138.0 kV: 111.0% loading (878 MVA) following the outage of the Southtown to Vienna Operations 345.0 kV L; DFAX = 45.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tasey 345/138kV Auto	Replace transformer with higher-rated unit	\$30,195,000	23% (\$6,946,000)
Vienna Operations 345 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$17,332,000	9% (\$1,502,000)
Sunet to Saddlebrooke Ranch Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$11,958,000	7% (\$840,000)
Chemolite to Randall County 138.0 kV L	Rebuild line with higher-capacity conductor	\$7,101,000	7% (\$486,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Newmark to Ess Lewes 138.0 kV L	Rebuild line with higher-capacity conductor	\$12,784,000	7% (\$899,000)
Galvanizing to Santeetlah Substation 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,585,000	7% (\$306,000)
Santeetlah to Sunset 138.0 kV	Rebuild line with higher-capacity conductor	\$5,351,000	7% (\$357,000)
Cady to Easton 2 345.0 kV Line	Rebuild line with higher-capacity conductor	\$71,257,000	7% (\$5,101,000)
Ripple 345 kV/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$16,355,000	7% (\$1,218,000)
Sandburg to Jarratt 138.0 kV L	Rebuild line with higher-capacity conductor	\$6,544,000	7% (\$477,000)
Sherwood to Sandburg 138.0 kV	Rebuild line with higher-capacity conductor	\$4,904,000	7% (\$358,000)
Hitchland to Sherwood 138.0 kV Line	Rebuild line with higher-capacity conductor	\$7,682,000	7% (\$560,000)
North Lake to Hitchland 138.0 kV L	Rebuild line with higher-capacity conductor	\$7,584,000	7% (\$553,000)
Kiwi to North Lake 138.0 kV	Rebuild line with higher-capacity conductor	\$4,462,000	7% (\$331,000)
Sunset to St Bonifacius 138.0 kV	Rebuild line with higher-capacity conductor	\$4,014,000	7% (\$284,000)
W -Station to St Bonifacius 138.0 kV line	Rebuild line with higher-capacity conductor	\$9,164,000	7% (\$649,000)
Saddlebrooke Ranch to Hansford 138.0 kV	Rebuild line with higher-capacity conductor	\$8,099,000	6% (\$480,000)
Quince Orchard to Southtown 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,492,000	4% (\$112,000)
Prunedale to Dane 138.0 kV	Rebuild line with higher-capacity conductor	\$3,495,000	4% (\$157,000)
Mesquite to Ripple 138.0 kV	Rebuild line with higher-capacity conductor	\$2,784,000	8% (\$231,000)
Mesquite to Arroyo Tap 138.0 kV	Rebuild line with higher-capacity conductor	\$2,146,000	8% (\$178,000)
Bristol to Limon Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$3,943,000	7% (\$266,000)
Grasonville to Wye Mills 138.0 kV	Rebuild line with higher-capacity conductor	\$16,462,000	8% (\$1,283,000)
Ocean City Substation to Reidland 138.0 kV	Rebuild line with higher-capacity conductor	\$5,546,000	8% (\$460,000)
Arroyo Tap to Reidland 138.0 kV	Rebuild line with higher-capacity conductor	\$2,247,000	8% (\$186,000)
	Total allocated to project		\$24,220,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.